# Exercise 3: Correlation filter tracking

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## I. Introduction

In this assignment we implemented a correlation filter tracker and evaluated its performance on the VOT2014 dataset. The tracker operates by learning a filter in the Fourier domain that produces a Gaussian peak response at the target's location. The filter is then updated over time using a learning rate to adapt to changes in the target's appearance and location. We then evaluated the tracker's performance using VOT-lite toolkit that uses standard performance measures accuracy and robustness.

## II. Experiments

The tracker was implemented using a simplified MOSS approach.

### A. Analysis of parameters

We analyzed how the learning rate $\alpha$ and the gaussian kernel bandwidth $\sigma$ influence the tracker's performance. Testing the tracker on the different sequences from the VOT2014 dataset, we found that a high learning rate (close to 1) allows the tracker to quickly adapt to changes in the targets appearance, but it can lead to instability, as it makes it more sensitive to occlusions (TODO woman example). Our thought was that a higher learning rate would then work best on the sequence with more dynamic targets (like in bolt and gymnastic). However, when tested on such sequences, we found that for all a learning rate at around 0.1 to 0.3 gives the best performance, when it comes to average overlap between the predicted and the ground truth bounding boxes. (TODO: add plots for all three sequences of the avg overlap). As seen in figure (TODO add) the graph is reverse U shaped, with the best performance close to the middle.

Focusing on the kernel bandwidth $\sigma$

### B. Larger patch

With the idea of capturing more background for better filter learning, we tested the tracker with patches larger than the target. So we extracted patches that were enlarged by some factor and then trained the filter on those. But the reported region (so the bounding box that the tracker returns) was still the same as the target size. Just to calculate the new center we used a larger window. This approach sometimes improved the performance, TODO especially when the target was moving fast and changing its appearance (gymnastics?). But a larger patch also made the tracker more sensitive to occlusions and background clutter, which could lead to more tracking failures. TODO add example of failures with larger patches.

### C. Tracking speed

Using the VOT-lite toolkit, we tested our tracker on the VOT2014 dataset and got the results seen in table I.

| | |
|---|---|
| Average overlap | 0.47 |
| Total failures | 66 |
| Average speed (FPS) | 814.30 |
| Total frames | 10213 |

Table I: Tracker results on the VOT2014 dataset.

## III. Conclusion